Work in progress — draft, not peer reviewed

A Lean 4 Formalization of the Classical Continuation Chain for Periodic Navier–Stokes on $\mathbb{T}^3$: Energy Identity, H^3 Commutator Reduction and the BKM Route

Companion paper to `BenFrohman/periodic-ns-classical-continuation`

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How to cite / What is claimed. In Lean 4 with Mathlib, we give a machine-checked reduction of Clay Millennium Statement (B) (periodic Navier–Stokes on $\mathbb{T}^3$) to the single hypothesis `PeriodicBKMIntegrability`. We prove this hypothesis is equivalent to global existence of smooth solutions for every smooth periodic initial field. The BKM criterion and H^3 control are proved unconditionally, using only the standard axioms. `PeriodicBKMIntegrability` remains open. The equivalence of vorticity integrability with continuation of smooth solutions is classical (Beale–Kato–Majda, 1984 [1]); the novelty here is the complete, `sorry`-free formalization. Cite as: B. S. Frohman, reduction paper, DOI 10.5281/zenodo.23842296; main paper, DOI 10.5281/zenodo.23842556.

Draft — tracks build in progress. This document reports exactly what the Lean development proves at the stated commit. It does *not* solve the Clay Millennium problem, and does not prove global regularity of the 3D Navier–Stokes equations. The step `PeriodicBKMIntegrability` is the open Clay core and remains an explicit hypothesis.

Abstract

We describe a Lean 4/Mathlib [4, 12] formalization of the classical continuation steps for smooth divergence-free solutions of the incompressible Navier–Stokes equations on the unit torus $\mathbb{T}^3$, along the Statement (B) route [5] of the companion repository `fefferman-clay-navier-stokes`. Proved with no hypothesis Props: local existence for $\nu \geq 0$, uniqueness via an L^2 energy-difference estimate, the Fourier coefficient ODE, coefficient decay, the H^s energy identity, Parseval's identity on the unit cube, and the Kato–Ponce toolkit [7] at $s = 3$ (integration by parts, Leibniz rule, a Gagliardo–Nirenberg inequality, a bound on every lower-order term, and vanishing of the top-order term from $\operatorname{div} u = 0$). Also proved: the $s = 3$ physical-space commutator bound `periodicPhysicalCommutatorBound`. Also proved: the low-frequency vorticity bound `periodicLowFreqVorticityBound`, and hence, with no hypothesis Props, H^3 Beale–Kato–Majda control (`periodicBKMHsControl_three`) and the periodic BKM criterion (`periodicBKMCriterion`). Statement (B) is proven conditionally on `PeriodicBKMIntegrability`, which is open (`clay_B_exact_of_integrability`). Small zero-mean H^1 data give $T^* = \infty$ conditionally on `PeriodicEnstrophyRateBound` and `PeriodicH1BlowupCriterion`; at $\kappa = 0$ the tether headline is equivalent to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_bending` gives integrability from `PeriodicDirectionEnstrophyGrowthL1`, `PeriodicH1BlowupCriterion` and the open `PeriodicBendingRateHiddenProperty`.

2020 Mathematics Subject Classification. 35Q30, 76D05, 68V15; 35B65, 03B35.

Keywords. Navier–Stokes equations; periodic domain; Beale–Kato–Majda criterion; continuation; Lean 4; Mathlib; formal verification; work in progress.

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1 Conventions and naming

Every hypothesis is a named `Prop` passed as an explicit theorem argument; no `axiom`, `sorry` or `admit` appears. The axiom audit (`lake env lean AxiomAudit.lean`) reports only `propext`, `Classical.choice` and `Quot.sound`. Lean identifiers defined in this repository are *coined in this build* (marked †), not names from the author's May/June 2026 manuscript. The identifiers `Clay_B_exact`, `PeriodicLocalExistence`, `PeriodicBKMCriterion` and `PeriodicBKMIntegrability` come from the dependency `fefferman-clay-navier-stokes` (pinned at `ebcf109`).

2 Relation to the May/June 2026 manuscript

The author's manuscript frames regularity through a tethered Lyapunov functional S_ε and a comparison ODE

$$y' = Cy^2 - \kappa'' y^3, \tag{2.1}$$

with transport cancellation, viscous dissipation, a Hölder–Sobolev gate and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). For *unmodified* Navier–Stokes ($\kappa = 0$) the cubic damping is absent, and the remaining gap is hypothesis (H5): a non-circular majorant for the vorticity. In this repository that gap appears exactly as `PeriodicBKMIntegrability`, i.e.

$$\int_0^T \|\omega(t)\|_{L^\infty}\,\mathrm{d}t < \infty. \tag{2.2}$$

Nothing here closes it. The transport cancellation of the manuscript corresponds to `top_term_zero` below.

3 Proved results (no hypothesis Props)

Theorem 3.1 (Local existence)**.** `periodicLocalExistence_of_nonneg`†*: for $\nu \geq 0$,* `PeriodicLocalExistence` *ν holds (Galerkin method at $s = 3$).*

Theorem 3.2 (Energy difference / uniqueness)**.** `periodicEnergyDifferenceEstimate`†*: for $\nu \geq 0$ and two classical solutions u, v with equal data,*

$$E(t) = \int_{[0,1]^3} |u - v|^2\,\mathrm{d}x \quad \textit{satisfies} \quad E'(t) \leq 2C\,E(t).$$

With Grönwall (`eq_zero_of_deriv_le_mul`†*,* `periodicVelocityUniqueness_of_energy`†*) this gives uniqueness.*

Theorem 3.3 (Fourier coefficient ODE, decay, H^s energy identity)**.** `periodicFourierCoeffODE`†*,* `periodicFourierCoeffDecay`† *and* `periodicFourierHsEnergyIdentity`† *(file* `CoeffODE.lean`*) prove the Props* `PeriodicFourierCoeffODE`*,* `PeriodicFourierCoeffDecay`*,* `PeriodicFourierHsEnergyIdentity` *for every ν (and every s): each Fourier coefficient $\hat{u}_k(t)$ obeys the projected ODE, coefficients decay faster than any power, and*

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{H^s}^2 = -2\nu \sum_k (1 + |k|^2)^s\, 4\pi^2 |k|^2 |\hat{u}_k|^2 + \big(\textit{nonlinear rate}\big). \tag{3.1}$$

The bridge is `periodicFourierHsEnergyIdentity_of_coeff`† *(*`HsEnergyIdentity.lean`*).*

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Theorem 3.4 (Parseval on the unit cube)**.** `hasSum_prod_cubeCoeff`† *(*`Parseval.lean`*): for continuous periodic f, g,*

$$\sum_{k\in\mathbb{Z}^3} \overline{\hat{f}_k}\,\hat{g}_k = \int_{[0,1]^3} \bar{f}\, g \,\mathrm{d}x,$$

via `torusLift`† *and Mathlib's torus Parseval.*

Theorem 3.5 (Physical form of the H^3 rate)**.** `hasSum_rate_three`†*: the Fourier-side H^3 nonlinear rate is summable and equals*

$$-2\sum_i \int \Big(1-\frac{\Delta}{4\pi^2}\Big)^3 v_i \left((v\cdot\nabla)v\right)_i \mathrm{d}x.$$

Theorem 3.6 (Kato–Ponce toolkit, $s=3$; `KatoPonce.lean`)**.** *Proved: unit-cell integration by parts* `cellI_ibp`†, `cellI_ibp'`†*; Leibniz rule* `DL_mul`†*; Gagliardo–Nirenberg on the cell* `gagliardoNirenberg_cell`† *and* `cellI_sq_mul_abs_le`†*; and*

- `lower_term_bound`†*: every lower-order Leibniz term (first factor of order ≥ 1) satisfies*

$$\left|\int D^L v_i \cdot D^{p_1} v_l \, D^{p_2}\partial_l v_i \,\mathrm{d}x\right| \leq 5\|\nabla v\|_{L^\infty}\, Q(v),$$

  with $Q =$ `physQ`†*;*

- `top_term_zero`†*:* $\displaystyle\sum_{i,l}\int D^L v_i\, v_l\, D^L\partial_l v_i \,\mathrm{d}x = 0$*, because* $\operatorname{div} v = 0$ *(transport cancellation).*

Theorem 3.7 (Continuation alternative)**.** `periodicBKMCriterion_of_hsControl`†*: for $\nu \geq 0$, $s > 5/2$,* `PeriodicBKMHsControl` *ν s implies* `PeriodicBKMCriterion` *ν (the H^s blow-up alternative is proved).*

4 Conditional results

Named hypothesis 4.1 (`PeriodicPhysicalCommutatorBound` C†, $s=3$)**.** For smooth periodic divergence-free v,

$$-2\sum_i \int \Big(1-\frac{\Delta}{4\pi^2}\Big)^3 v_i \left((v\cdot\nabla)v\right)_i \mathrm{d}x \leq C\|\nabla v\|_{L^\infty}\|v\|_{H^3}^2. \tag{4.1}$$

Status: **proved** in `CommutatorAssembly.lean`: `periodicPhysicalCommutatorBound`† gives it with $C = 2\cdot 64\cdot 315\cdot 64\,(4\pi^2)^3$, from `lower_term_bound`, `top_term_zero`, `word_bound`† and `physQ_le_hsSq`†. No longer a hypothesis.

Named hypothesis 4.2 (`PeriodicLowFreqVorticityBound`†)**.** There is $K \geq 0$ such that for all $R \geq 2$,

$$\|V_R \nabla v\|_{L^\infty} \leq K(1+\log R)\|\omega\|_{L^\infty}. \tag{4.2}$$

Status: **proved** in `LowFreqAssembly.lean` (`periodicLowFreqVorticityBound`†). The proof writes the Biot–Savart law in Riesz form,

$$\widehat{\partial_j v_i} = q_{j,i+2}\,\hat{\omega}_{i+1} - q_{j,i+1}\,\hat{\omega}_{i+2}, \qquad q_{jb} = \frac{k_j k_b}{|k|^2}.$$

It splits q into a low piece and $S = \lceil \log_2 R\rceil + 3$ smooth dyadic blocks, each with a uniformly L^1-bounded kernel (`DyadicBlock.lean`). Every piece is followed by the uniformly L^1-bounded de la Vallée Poussin kernel (`VPKernel.lean`). No longer a hypothesis.

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Named hypothesis 4.3 (`PeriodicBKMIntegrability` ν)**.** From the dependency `fefferman-clay-navier-stokes`: $\int_0^T \|\omega\|_{L^\infty}\,\mathrm{d}t < \infty$ on every finite interval of existence. *Status:* **open**. This is the Clay core and the manuscript's (H5) gap.

Proposition 4.4. `periodicFourierCommutatorBound_of_physical`†*:* `PeriodicPhysicalCommutatorBound` $C \Rightarrow$ `PeriodicFourierCommutatorBound` 3 C*.* `periodicBKMHsControl_of_physical_lowFreq`†*:* ($\exists C \geq 0$, `PeriodicPhysicalCommutatorBound` C) *and* `PeriodicLowFreqVorticityBound` *give* `PeriodicBKMHsControl` ν 3*. Combined with the proved bound:* `periodicBKMHsControl_three_of_lowFreq`† *gives* `PeriodicBKMHsControl` ν 3 *from* `PeriodicLowFreqVorticityBound` *alone.*

Theorem 4.5 (Unconditional corollaries, `BKMFinal.lean`)**.** `periodicBKMHsControl_three`†*:* `PeriodicBKMHsControl` ν 3 *for every* ν*, with no hypotheses.* `periodicBKMCriterion`†*:* `PeriodicBKMCriterion` ν *for every* ν*, with no hypotheses.*

Proposition 4.6 (Statement (B), conditional)**.** `clay_B_exact_of_hsControl`†*: for* $s > 5/2$*,* `PeriodicBKMHsControl` ν s *and* `PeriodicBKMIntegrability` ν *give* `Clay_B_exact` ν*. Also* `clay_B_exact_of_corrected_route'`† *and* `clay_B_exact_of_bkm`†*. The final form,* `clay_B_exact_of_integrability`†*, is conditional* only *on the open* `PeriodicBKMIntegrability` ν*.*

Remark 4.7 (A failed route). A Wiener-algebra/Fourier commutator estimate was tried and abandoned: its low-frequency part grows like $R^{3/2}$, not $\log R$. The $s = 3$ physical-space estimate replaces it.

5 Attacking the open hypothesis (work in progress)

Files `IntegrabilityAttack.lean`, `SmallDataAttack.lean`, `TetherClaimAttack.lean`, `EnstrophyIdentity.lean`, `TetherKappaZero.lean`, `DirectionCriterion.lean` (commits `f981e71`, `052cef1`, `ea5133d`, `0cd3551`, `a35d13a`). All names are coined. **This does not prove `PeriodicBKMIntegrability`.**

- `integrableOn_vorticity_of_lt_tstar` (proved): if $T < T^*$, then $\int_0^T \|\omega\|_{L^\infty}\,\mathrm{d}t < \infty$.
- `periodicBKMIntegrability_iff_globalLifespan` (proved): `PeriodicBKMIntegrability` ν holds iff `PeriodicGlobalLifespan` ν ($T^* = \infty$ for all smooth data). The open hypothesis is therefore exactly global existence.
- Route 2 (proved from named classical hypotheses, not formalized here):
  - `integrability_of_smallDataGlobal`, from `PeriodicSmallDataGlobal` [6];
  - `integrability_of_prodiSerrin`, from `PeriodicProdiSerrinCriterion` [10, 11] (endpoint $L^\infty_t L^\infty_x$) plus a velocity bound.
- Route 1 (the $\kappa = 0$ analogue of the tether), `periodicBKMIntegrability_of_vorticityAPriori` (proved): from the **open** `PeriodicVorticityAPrioriBound`. It breaks because with no damping term the enstrophy satisfies only $E' \leq CE^3$ and the max-point bound gives $Y' \lesssim Y^2 \log(\cdot)$. Neither gives a uniform bound.
- `periodicEnstrophyIdentity` (`EnstrophyIdentity.lean`, proved, **no hypotheses**): the $\mathbb{T}^3$ enstrophy (H^1) energy identity, the $s = 1$ case of the H^s identity (3.1).
- Small zero-mean H^1 data (`EnstrophyIdentity.lean`): `riccati_barrier_Ioo` (proved) and `periodicTstar_eq_top_of_smallH1_of_rate`, proved from the unconditional identity plus the named classical hypotheses `PeriodicEnstrophyRateBound` (Ladyzhenskaya/Poincaré trilinear bound [8]) and `PeriodicH1BlowupCriterion` (Leray [9]), both

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not formalized at the time. *Update:* the trilinear bound `PeriodicH1TrilinearBound` is now proved (`periodicH1TrilinearBound`, `Ladyzhenskaya.lean`, explicit unoptimized constant, Fourier/Agmon route to a Ladyzhenskaya-type estimate; the inequality is classical, O. A. Ladyzhenskaya 1958 in 2D and her 1961 monograph in 3D [8]), as is `PeriodicMeanConservation` (`periodicMeanConservation`); hence `periodicTstar_eq_top_of_smallH1_ladyzhenskaya` depends only on the named `PeriodicH1BlowupCriterion`, which remains **not formalized**. This does not resolve the Clay problem. *Bug fix:* the earlier hypothesis `PeriodicEnstrophyRiccati` with $\alpha > 0$ (`SmallDataAttack.lean`, used by `periodicTstar_eq_top_of_smallH1`, `integrability_of_smallH1`) is false for small constant flows, since the mean mode is conserved; the zero-mean rate-bound version replaces it.

- Tether claim (`TetherClaimAttack.lean`): `cubic_riccati_barrier` (proved); `periodicVorticityAPrioriBound_of_tetherCubicRate`, `clay_B_exact_of_tetherCubicRate` proved from the **open** `PeriodicTetherCubicRate` (not derived from unmodified NS); `uniform_cubic_rate_false_of_scaling` (proved, abstract scaling step: a data-uniform cubic rate is contradictory).

- Direction route: `integrability_of_directionCriterion`, `periodicBKMIntegrability_of_direction`, proved from the classical `PeriodicDirectionCriterion` (Beirão da Veiga–Berselli [2] / Constantin–Fefferman [3], not formalized) plus the **open** `PeriodicDirectionRegularity`. New: `periodicDirectionCriterion_of_growth` proves `PeriodicDirectionCriterion` from the unconditional enstrophy identity, the linear Grönwall lemma `exp_gronwall_Ioo`, and two named open hypotheses: `PeriodicDirectionEnstrophyGrowth` (coined; Constantin–Fefferman 1993 geometric depletion of the strain kernel plus Beirão da Veiga–Berselli 2002, constant-rate form, not formalized) and `PeriodicH1BlowupCriterion` (Leray 1934). `periodicBKMIntegrability_of_directionGrowth` still needs the open `PeriodicDirectionRegularity`.

- Bending rate (`BendingRate.lean`): `hasDerivAt_vortDir_lagrangian` (proved) gives the pointwise identity

$$\frac{D\mathbf{t}}{Dt} = (I - \mathbf{t}\otimes\mathbf{t})S\mathbf{t} + \nu|\omega|^{-1}(I - \mathbf{t}\otimes\mathbf{t})\Delta\omega, \qquad \mathbf{t} = \omega/|\omega|, \tag{5.1}$$

  for any curve with $D\omega/Dt = S\omega + \nu\Delta\omega$; `inner_perpPart` shows the orthogonal strain is invisible to the stretching rate $\mathbf{t}\cdot S\mathbf{t}$. The hidden-property hypothesis `PeriodicBendingRateHiddenProperty` (coined; new formulation dated Oct 31 2026; precursors (geometric only, no regularity claims): the author's May 17/21 2026 notes (Π_u orthogonal to u, "invisible to the dynamics generated by H") and Aug 10 2026 notes *The Frohmanian Symplectic Tether* (B_{anti} via $\hat{u}\times(\cdot)$, B_{sym} with pointwise Π_u)) is the conjunction of `PeriodicOrthogonalStrainControl` and `OrthogonalStrainToDirectionHolder`, both **open**; `periodicDirectionRegularity_of_bending` (proved) derives `PeriodicDirectionRegularity` from it. Classical step broken down: `abs_cfDepletion_le` (proved) bounds the Constantin–Fefferman kernel factor $(e\cdot a)\det(e,a,b)$ by $|b-a|$, and `cfDepletion_le_of_holder` by $K|x-y|^{1/2}$; `gronwall_integrating_factor` (proved) is Grönwall with rate $g = G' \in L^1$; `periodicDirectionCriterion_of_growthL1` (proved) gives `PeriodicDirectionCriterion` from the weaker open `PeriodicDirectionEnstrophyGrowthL1` (Biot–Savart stretching representation on $\mathbb{T}^3$ + BdV–B interpolation, not formalized) and `PeriodicH1BlowupCriterion`.

- Leray and analytic CF breakdown (`ClassicalSteps.lean`): `hsSq_bounded_of_rate` (proved; linear rate bound ⇒ bounded $\|u\|_{H^s}^2$ on $[0,T^*)$); `periodicH1BlowupCriterio`

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`n_of_smoothing` (proved) derives `PeriodicH1BlowupCriterion` from the open parabolic-smoothing rates `PeriodicH2SmoothingRate` ($H^1 \to H^2$) and `PeriodicH3SmoothingRate` ($H^2 \to H^3$) together with the proved H^3 blow-up alternative; `periodicDirectionEnstrophyGrowthL1_of_holderStretching` (proved) derives `PeriodicDirectionEnstrophyGrowthL1` from the open `PeriodicHolderStretchingEstimate` (Biot–Savart stretching on $\mathbb{T}^3$ + BdV–B) and `PeriodicEnstrophyTimeIntegrable` (energy equality); `periodicBKMIntegrability_of_classical_bending` (conditional).

- $\kappa = 0$ tether (`TetherKappaZero.lean`): `tetherForm_zero`, `tetherGeometricControl_zero`, `tetherImpliesDirection_zero_iff` (proved): at $\kappa = 0$ the tether control is vacuous, so "tether control ⇒ direction regularity" is *equivalent* to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_tetherKappaZero` (conditional).

6 Status

- **Proved, unconditional:** local existence ($\nu \geq 0$), uniqueness, coefficient ODE and decay, H^s energy identity, Parseval, physical form of the H^3 rate, Kato–Ponce toolkit including `lower_term_bound` and `top_term_zero`, the $s = 3$ commutator bound `periodicPhysicalCommutatorBound`, the blow-up alternative, the low-frequency bound `periodicLowFreqVorticityBound`, H^3 BKM control `periodicBKMHsControl_three` and the BKM criterion `periodicBKMCriterion`, the equivalence `periodicBKMIntegrability_iff_globalLifespan`, and the H^1 enstrophy identity `periodicEnstrophyIdentity`.
- **Conditional:** Statement (B) is proven conditionally on `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). Small zero-mean H^1 data give $T^* = \infty$ conditionally on `PeriodicEnstrophyRateBound` and `PeriodicH1BlowupCriterion`; at $\kappa = 0$ the tether headline is equivalent to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_bending` gives integrability from `PeriodicDirectionEnstrophyGrowthL1`, `PeriodicH1BlowupCriterion` and the open `PeriodicBendingRateHiddenProperty`.
- **Direction route, L^1 form:** `periodicDirectionCriterion_of_growthL1` gives `PeriodicDirectionCriterion` from `PeriodicDirectionEnstrophyGrowthL1` and `PeriodicH1BlowupCriterion`; these in turn reduce (proved) to `PeriodicHolderStretchingEstimate`, `PeriodicEnstrophyTimeIntegrable`, `PeriodicH2SmoothingRate`, `PeriodicH3SmoothingRate`, all open here. The bending-rate hypothesis `PeriodicBendingRateHiddenProperty` (coined Oct 31 2026, open) implies `PeriodicDirectionRegularity`; the identity $D\mathbf{t}/Dt = (I - \mathbf{t} \otimes \mathbf{t})S\mathbf{t} + \nu|\omega|^{-1}(I - \mathbf{t} \otimes \mathbf{t})\Delta\omega$ is proved.
- **Open:** `PeriodicBKMIntegrability` — the Clay Millennium core. Neither the Clay problem nor global regularity of 3D Navier–Stokes is claimed.

Reproducibility

`lake build`; `lake env lean AxiomAudit.lean` (720 theorems and lemmas in this repository; 0 `sorry`; only `propext`, `Classical.choice`, `Quot.sound`).

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